

North Sea Oil Platform Drill Tower Construction

Document ID: EVM-COMPREHENSIVE-REPORT-2026-M08

Status Date: August 31, 2026

Author: Frank Ellingsen, Lead Project Controller

1. Executive Steering Committee Status Briefing

Project Name: Offshore EPC Platform Drill Tower Project

Status Date: August 31, 2026 (Month 8 / Status Week 36)

Report Type: Monthly Executive Steering Committee Status Briefing

Author: Frank Ellingsen, Lead Project Controller

OVERALL PROJECT HEALTH DASHBOARD: RED (CRITICAL)

Overall RAG Status	Cost Performance (\$CPI\$)	Schedule Performance (\$SPI_t\$)	Budget Burn Out Month	Predicted Outturn Cost (\$EAC_1\$)	Predicted Commercial COD
 CRITICAL OVERRUN	0.7483 (Red)	0.9250 (Amber)	Month 9 (Sep 2026)	\$35,413,604 (+\$8.91M)	Jan 31, 2027 (+31 Days)

1. KEY PERFORMANCE INDICATORS & EVM METRICS SUMMARY

- Baseline Budget (\$BAC\$):** \$26,500,000
- Earned Physical Completion (\$EV / BAC\$):** 66.19% (\$EV = \$17,540,000)
- Planned Baseline Completion (\$PV / BAC\$):** 77.36% (\$PV = \$20,500,000)
- Actual Costs Incurred (\$AC\$):** \$23,440,000
- Cost Variance (\$CV = EV - AC\$):** -\$5,900,000 (-33.6% cost overrun to date)
- Schedule Variance (\$SV = EV - PV\$):** -\$2,960,000 (-14.4% work lag to date)
- Earned Schedule (\$ES\$):** 7.40 Months (\$SV_t = -18.2 Days, \$SPI_t = 0.9250\$)

2. OUTTURN FORECASTS & MONTE CARLO P90 RISK ANALYSIS

Outturn Scenario	Forecast Cost (\$EAC\$)	Cost Overrun (\$VAC\$)	Completion Date	Schedule Delay	Required Risk Contingency
Deterministic Most Likely (\$EAC_1\$)	\$35,413,604	-\$8,913,604	Jan 31, 2027	+31 Days	Base Forecast (\$0.00)
Monte Carlo P50 (Median)	\$34,060,783	-\$7,560,783	Feb 27, 2027	+57 Days	Base Forecast (\$0.00)
Monte Carlo P90 (90% High Confidence)	\$35,815,202	-\$9,315,202	Mar 14, 2027	+72 Days	+\$401,598 Reserve

3. PRIMARY COST & SCHEDULE VARIANCE DRIVERS

- Egersund Mast Assembly Yard Rework (\$WBS\text{ }1.3.2\$):** Structural pipe out-of-tolerance requiring 24/7 NDT welding. Accounts for **\$2.40M direct overrun** (26.9% of total variance).
- Time Delay Overhead Spread (\$SPI_t = 0.9250\$):** Extending project execution past Dec 31, 2026 adds **\$3.81M** in time-based site overhead (42.8% of total variance).

3. **Heavy Lift Vessel Weather Standby (\$WBS\text{ }1.4.1\$)**: Marine vessel mobilization delayed into autumn sea states (*Heerema Sleipnir*), incurring **\$1.30M standby cost** (\$14.6\%\$ of variance).

4. CASH RUNWAY & LIQUIDITY ALERT

- **Remaining Baseline Capital (\$BAC - AC\$)**: **\$3,060,000**
- **Average Monthly Cash Burn Rate**: **\$2,930,000 / month**
- **Budget Burn Out Date**: **Month 9 (September 2026)** — 100% of the \$26.50M baseline budget will be depleted.
- **Urgent Action**: Secure executive approval for a **+\$8.91M overrun credit line** before September 15, 2026.

5. RECOMMENDED STEERING COMMITTEE DECISIONS

1. **Approve Additional Funding Line**: Authorize **\$8.91M** overrun credit line to cover Months 9–13 operations.
2. **Implement Fixed-Fee Yard Agreement**: Cap Egersund yard billing under a fixed-fee labor agreement for remaining assembly.
3. **Execute Critical Path Recovery**: Shift 15 calendar days from hook-up testing to preserve **January 31, 2027 COD**.

2. Earned Value & Outturn Predictions

****Project Name**:** Offshore EPC Platform Drill Tower Project

****Status Date**:** August 31, 2026 (Month 8 / Status Week 36)

****Author**:** Frank Ellingsen, Lead Project Controller

Executive Summary & Outturn Prediction Matrix

At Status Month 8, the project has achieved ****66.19% physical completion**** ($EV = \$17.54\text{M}$) against a cumulative budget spent of ****\$23.44M**** (\$AC\$) and a baseline plan of ****\$20.50M**** (\$PV\$).

Based on standard Earned Value Management (PMI EVM Practice Standard) and Earned Schedule (\$ES\$) statistical modeling, the project ****will not achieve its original \$26.50M budget or December 31, 2026 completion date****.

Summary of Final Outcome Scenarios

Forecast Scenario	Metric	Outturn Cost (\$EAC\$)	Variance (\$VAC\$)	Completion Date	Schedule Delay (\$TV\$)	Probability	Primary Driving Assumptions
Best Case (Idealistic)	\$EAC_2\$	**\$32.40M**	-\$5.90M (-22.3%)	**Jan 15, 2027**	+15 Days	5%	Future work performed strictly at 100% budget efficiency ($CPI_{\text{future}} = 1.00$).
Most Likely (Standard EVM)	\$EAC_1\$	**\$35.41M**	**-\$8.91M (-33.6%)**	**Jan 31, 2027**	**+31 Days**	**70%**	**Remaining work performed at current cumulative efficiency ($CPI = 0.75$, $SPI_t = 0.925$).**
Worst Case (Composite)	\$EAC_3\$	**\$37.43M**	-\$10.93M (-41.3%)	**Feb 15, 2027**	+46 Days	25%	Cumulative cost & schedule compounding ($CPI \times SPI = 0.64$).

1. Cost Predictions (\$EAC\$ & \$VAC\$)

Status Metrics at Month 8 (\$M8\$)

- **Budget at Completion (\$BAC\$)**:** $\$26,500,000$
- **Planned Value (\$PV\$)**:** $\$20,500,000$
- **Earned Value (\$EV\$)**:** $\$17,540,000$ (66.19% physical complete)
- **Actual Cost (\$AC\$)**:** $\$23,440,000$
- **Cost Performance Index (\$CPI\$)**:** 0.7483 (-25.2% cost efficiency)

Detailed \$EAC\$ Calculations

1. ****Standard EVM Forecast (\$EAC_1\$ - Most Likely)**:**

$$EAC_1 = \frac{BAC}{CPI} = \frac{\$26,500,000}{0.74829} = \$35,413,604$$

- **Cost Overrun (\$VAC_1\$)**:** $-\$8,913,604$ (+33.6% cost overrun).
- **Root Cause**:** Unbudgeted subsea structural steel rework ($WBS_{\text{}}1.3.2$) and crane barge standby rates ($WBS_{\text{}}1.4.1$).

2. ****Floor / Baseline Forecast (\$EAC_2\$ - Best Case)**:**

$$EAC_2 = AC + (BAC - EV) = \$23,440,000 + \$8,960,000 = \$32,400,000$$

- **Cost Overrun (\$VAC_2\$)**:** $-\$5,900,000$ (Freeze current overrun; zero future variance).

3. ****Composite Risk Forecast (\$EAC_3\$ - Worst Case)**:**

$$EAC_3 = AC + \frac{BAC - EV}{CPI \times SPI} = \$23,440,000 + \frac{\$8,960,000}{0.7483 \times 0.8556} = \$37,433,522$$

- **Cost Overrun (\$VAC_3\$):** \$-10,933,522\$ (+41.3% cost overrun).

2. Schedule & Completion Date Predictions (\$EAC_t\$)

Time-Based Status Metrics (\$ES\$ Lipke Theory)

- **Actual Time (\$AT\$):** 8.00\text{ Months}\$ (240\text{ Days}\$)
- **Earned Schedule (\$ES\$):** 7.40\text{ Months}\$ (222\text{ Days}\$)
- **Schedule Performance Index (\$SPI_t\$):** 0.9250\$ (-7.5% time efficiency)
- **Time Variance (\$SV_t\$):** -0.60\text{ Months}\$ \approx -18.2\text{ Days}\$

Completion Date Predictions

1. **Earned Schedule Forecast (\$EAC_t\$ - Time Estimate at Completion):**

$$EAC_t = \frac{PD}{SPI_t} = \frac{12.00\text{ Months}}{0.9250} = 12.97\text{ Months} \approx 13.0\text{ Months}$$

- **Forecast Finish Date:** January 31, 2027
- **Project Delay:** +31 Calendar Days\$ (1.0\text{ Month}\$)

2. **Critical Path Schedule Driving Logic:**

- **Driving Task:** WBS\text{ }1.3.2\$ Derrick Tower Mast Assembly (Egersund Yard).
- **Constraint:** Heavy Lift Vessel (*Heerema Sleipnir*) mobilization window is locked from Sep 15 to Oct 15.
- **Cascade Risk:** A 30-day delay in mast assembly shifts vessel mobilization into Oct 15–Nov 15, exposing offshore installation to North Sea winter weather downtime (+15-20 days severe weather standby risk).

3. To-Complete Performance Index (\$TCPI\$) Feasibility Analysis

Baseline Metric	Target	Required Efficiency (\$TCPI\$)	Feasibility Evaluation	Operational Action
Original Budget (\$BAC\$)	\$26.50\text{M}\$	\$2.93\$ (or \$6.07\$ peak)	UNVAILABLE	Impossible to achieve. Abandon \$BAC\$ as financial control baseline.
Revised Forecast (\$EAC_1\$)	\$35.41\text{M}\$	\$1.00\$ (\$0.75\$)	ACHIEVABLE	Standard target. Requires strict cost control on remaining \$8.96\text{M}\$ work.

4. Key Project Controller Recommendations

1. **Formal Baseline Revision (Re-baseline EAC):**

- Formally submit a Change Order to the Steering Committee updating the baseline \$BAC\$ from \$26.50M\$ to \$35.41M\$ and extending target completion to January 31, 2027.

2. **Yard Rework Clampdown at Egersund (\$WBS\text{ }1.3.2\$):**

- Issue a fixed-fee cap on Egersund assembly yard labor. Require 2-shift operations (16 hrs/day) to complete mast assembly by September 30.

3. **Heerema Vessel Standby Mitigation (\$WBS\text{ }1.4.1\$):**

- Re-negotiate Heavy Lift Mobilization window with Heerema Marine Contractors to avoid \$150,000/day standby penalties during weather delays.

4. **Bi-Weekly Earned Schedule (\$ES\$) Tracking:**

- Track \$SPI_t\$ bi-weekly. If \$SPI_t\$ drops below 0.90\$, immediately trigger emergency double-shift welding crews.

3. Monte Carlo Schedule & Cost Risk Analysis (P50 / P90)

****Project Name**:** Offshore EPC Platform Drill Tower Project

****Simulation Iterations**:** 10,000 Runs (Triangular Risk Sampling)

****Status Date**:** August 31, 2026 (Month 8 / Status Week 36)

****Author**:** Frank Ellingsen, Lead Project Controller

Executive Summary & P90 Risk Card

A quantitative cost and schedule risk analysis (QCRA / QSRA) was executed using 10,000 Monte Carlo iterations across all 7 Control Accounts and critical path schedule deliverables.

> [!IMPORTANT]

> ****P90 Outturn Forecast Summary**:**

> - ****P90 Outturn Cost**:** ****\$35,815,202**** (\$-9.32\text{M}\$ / \$+35.2\%\$ variance vs. \$26.50\text{M}\$ baseline \$BAC\$).

> - ****P90 Contingency Reserve Needed**:** ****+\$401,598**** added to current \$35.41\text{M}\$ deterministic \$EAC\$.

> - ****P90 Completion Date**:** ****March 14, 2027**** (\$+72.5\text{ Days}\$ / \$+2.4\text{ Months}\$ schedule delay vs. Dec 31, 2026).

Summary Table of Monte Carlo Risk Percentiles

Risk Percentile	Confidence Level	Outturn Cost (\$EAC\$)	Cost Variance vs \$BAC\$	Contingency Needed vs \$EAC\$	Total Duration	Predicted Completion Date	Schedule Delay vs Dec 31
P10	10% (Optimistic / Low Risk)	**\$32,444,302**	-\$5,944,302 (-22.4%)	\$0.00	408.1 Days	**Feb 13, 2027**	+43.1 Days (+1.4 Mos)
P50	50% (Median / Expected)	**\$34,060,783**	-\$7,560,783 (-28.5%)	\$0.00	422.3 Days	**Feb 27, 2027**	+57.3 Days (+1.9 Mos)
P80	80% (Standard Budget Baseline)	**\$35,195,026**	-\$8,695,026 (-32.8%)	\$0.00	432.3 Days	**Mar 09, 2027**	+67.3 Days (+2.2 Mos)
P90	**90% (P90 High Confidence)**	**\$35,815,202**	**-\$9,315,202 (-35.2%)**	**+\$401,598**	**437.5 Days**	**March 14, 2027**	**+72.5 Days (+2.4 Mos)**
P95	95% (Extreme Risk Boundary)	**\$36,272,986**	-\$9,772,986 (-36.9%)	+\$859,382	441.6 Days	**March 18, 2027**	+76.6 Days (+2.5 Mos)

1. Simulation Methodology & Input Risk Distributions

The simulation applies triangular probability distributions \$(Optimistic, Most\text{Likely}, Pessimistic)\$ to each Control Account:

Cost Inputs (\$M USD)

- **WBS 1.1 Detail Engineering**:** \$(3.10, 3.50, 4.20)\$
- **WBS 1.2 Procurement**:** \$(8.20, 9.10, 10.80)\$
- **WBS 1.3.1 Verdal Fabrication**:** \$(4.80, 5.20, 6.00)\$
- **WBS 1.3.2 Egersund Mast Rework**:** \$(5.50, 7.20, 9.50)\$ $\times$ (Primary Cost Risk)*
- **WBS 1.4.1 Heavy Lift Vessel Mobilization**:** \$(3.20, 4.50, 6.80)\$ $\times$ (Primary Weather Standby Risk)*
- **WBS 1.4.2 Topside Lifting & Mating**:** \$(1.10, 1.40, 2.10)\$
- **WBS 1.5 Hook-up & Commissioning**:** \$(1.70, 1.90, 2.50)\$

Schedule Duration Inputs (Remaining Days along Critical Path)

- **T106 Derrick Mast Assembly (Egersund)**: \$(20, 30, 45\text{ Days})\$
- **T107 Heavy Lift Mobilization**: \$(25, 30, 50\text{ Days})\$
- **T108 Topside Lifting & Mating**: \$(25, 31, 45\text{ Days})\$
- **T109 Structural Hook-up & NDT**: \$(25, 30, 45\text{ Days})\$
- **T110 Pre-Commissioning & Handover**: \$(35, 47, 65\text{ Days})\$

2. Risk Driver Sensitivity & Tornado Analysis

1. **Egersund Assembly Yard Rework (\$WBS\text{ }1.3.2\$)**: Accounts for **48.2%** of total cost variance. Structural pipe tolerance issues require 24/7 NDT inspection and specialized welding crews.

2. **Offshore Weather Standby (\$WBS\text{ }1.4.1\$)**: Accounts for **36.5%** of total schedule variance. Shifting vessel mobilization from September into October/November exposes the heavy lift crane barge (*Heerema Sleipnir*) to North Sea sea-state limits (\$H_s > 2.5\text{m}\$).

3. Project Controller Recommendations for Steering Committee

1. **Establish P80 / P90 Financial Reserve**:

- Approve a revised Control Budget of **\$35.82M** (\$P90) by allocating a **+\$401,598** management contingency reserve on top of the **\$35.41\text{M}** deterministic \$EAC\$.

2. **Schedule Target & Buffer**:

- Establish **March 14, 2027** as the P90 contractual handover date for offshore commissioning.

3. **CSV Export**:

- Simulation raw dataset saved to [03_power_bi/Monte_Carlo_Simulation_Results.csv]
(file:///C:/Users/frank/Desktop/EVM/03_power_bi/Monte_Carlo_Simulation_Results.csv).

4. EVM Variance Explanations & Corrective Action Audit

Project: North Sea Oil Platform Drill Tower Construction (\$26.5M EPC Program)

Status Date: August 31, 2026 (Month 8 / Status Date)

Target Role: Project Controller (Frank Ellingsen) / Executive Steering Committee

Executive Summary & Red Indicator Audit

At Status Month 8, the North Sea Drill Tower program exhibits critical cost overruns (\$CPI = 0.75\$) and critical path schedule delays (\$+31\text{ Days}\$, \$SPI = 0.86\$). The table below outlines the root cause explanations and recommended corrective action plans for all active red performance indicators.

Red Indicator	Status Metric	Threshold	Root Cause Explanation	Recommended Actionable Next Steps	Accountable CAM
Cost Variance (\$CV\$)	-\$5,900,000	\$CPI = 0.75\$ (\$< 0.90\$)	Structural steel fitting rework, unbudgeted welding labor hours, and premium overtime rates at Verdal and Egersund yards.	Enforce Design Change Freeze on WBS 1.3 deliverables. Convert open T&M subcontracts to fixed unit-rate capped milestones.	O. Eriksen
Outturn Deficit (\$VAC\$)	-\$8,913,604	\$EAC = \$35.41\text{M}\$	Linear extrapolation of poor historical cost performance (\$CPI = 0.75\$) indicates 33.6% budget overrun against \$BAC\$.	Submit formal EAC re-baseline proposal to Executive Steering Committee. Request \$3.5\text{M}\$ drawdown from Management Reserve.	Frank Ellingsen
Target Efficiency (\$TCPI_{BAC}\$)	6.07 (BAC)	Unviable (\$> 1.10\$)	Original \$BAC\$ budget target requires impossible cost efficiency of 607% for remaining work.	Formally transition operational target to \$TCPI_{EAC} = 1.00\$. Enforce weekly EVM gate reviews for CAMs with \$TCPI_{EAC} > 1.05\$.	Frank Ellingsen
Schedule Slippage (\$SV\$)	-\$2,960,000	\$+31\text{ Days}\$ Delay	Procurement delay on Tubular Steel (T103) cascaded to Verdal (T105) and Egersund (T106), threatening Heavy Lift window.	Fast-track Egersund rigging with parallel work-fronts. Re-sequence pre-commissioning tasks. Negotiate 10-day vessel window extension.	M. Berg / K. Solberg
WBS 1.3.2 Mast Assembly	-\$2,330,000	\$CPI = 0.57\$	Egersund rigging yard productivity bottleneck; high sub-tier contractor churn and excessive rework hours.	Deploy resident Project Controller to Egersund full-time. Audit daily labor timecards and enforce weld quality gates.	O. Eriksen

Detailed Variance Analysis & Recommended Actions

1. Cost Variance (\$CV = -\\$5,900,000\$, \$CPI = 0.75\$)

- Root Cause Explanation:** Actual expenditure (\$AC = \$23.44\text{M}\$) severely exceeds Earned Value (\$EV = \$17.54\text{M}\$). Major cost growth originated in WBS 1.3.1 (Verdal Yard Sub-Structure Fabrication, \$CV = -\\$850\text{k}\$) and WBS 1.3.2 (Egersund Derrick Mast Assembly, \$CV = -\\$2.33\text{M}\$). Key drivers include engineering change notices during fabrication, out-of-sequence welding, and high overtime premiums.
- Recommended Action Plan:**

- Design Change Freeze:** Formally freeze all pending Engineering Change Requests (CRs) for WBS 1.3 until offshore installation completion.
- Subcontract Conversion:** Convert open Time & Materials (T&M) subcontracts with yard fabrication suppliers into fixed unit-rate capped milestones.

3. **On-site Controller Audit**: Station a dedicated Project Controller at Egersund yard to approve daily man-hour timecards and verify physical milestone completion before payment release.

2. **Outturn Forecast Deficit** (\$VAC = -\$8,913,604\$, \$EAC = \$35,413,604\$)

- Root Cause Explanation**: Extrapolating historical efficiency (\$CPI = 0.75\$) yields a cumulative \$EAC_{CPI}\$ of \$35.41\$ against the baseline \$BAC\$ of \$26.50\$, creating an outturn deficit of \$8.91\$.
- Recommended Action Plan**:
 - EAC Re-baseline Proposal**: Prepare an executive re-baseline submission for the Steering Committee to adjust the project budget baseline to \$35.41\$.
 - Management Reserve (MR) Drawdown**: Formally request a \$3.50\$ allocation from corporate Management Reserve to absorb non-recoverable raw material price inflation.
 - ETC Ceiling Enforcement**: Set strict Estimate-to-Complete (\$ETC = 11.97\$) expenditure caps per WBS control account.

3. **Target Cost Efficiency** (\$TCPI_{BAC} = 6.07\$)

- Root Cause Explanation**: \$TCPI_{BAC} = \frac{BAC - EV}{BAC - AC} = \frac{26.50 - 17.54}{26.50 - 23.44} = 6.07\$. Achieving the remaining work within the original \$26.50\$ budget is mathematically unviable, requiring \$607\%\$ cost efficiency.
- Recommended Action Plan**:
 - Pivot Operational Metric**: Cease using \$TCPI_{BAC}\$ as an operational performance target.
 - Enforce \$TCPI_{EAC}\$ Tracking**: Monitor remaining work against \$TCPI_{EAC} = \frac{BAC - EV}{EAC - AC} = \frac{26.50 - 17.54}{35.41 - 23.44} = 1.00\$.
 - Weekly CAM Reviews**: Implement mandatory weekly EVM variance gate reviews for any Control Account Manager whose \$TCPI_{EAC}\$ exceeds \$1.05\$.

4. **Critical Path Schedule Delay** (\$SV = -\$2,960,000\$, \$+31\$ Days\$)

- Root Cause Explanation**: Procurement delays on High-Grade Tubular Steel (T103, \$+31\$ days) delayed Verdal Sub-Structure Fabrication (T105) and Egersund Derrick Assembly (T106). This pushes Heavy Lift Mobilization (T107) into late September/October, risking autumn North Sea weather windows.
- Recommended Action Plan**:
 - Rigging Fast-Tracking**: Parallelize structural mast assembly work-fronts at Egersund by deploying additional heavy lift cranes.
 - Pre-Commissioning Re-sequencing**: Shift 14 days of pre-commissioning testing (T110) post-mating to shorten the critical path prior to vessel mobilization.
 - Marine Fleet Contingency**: Negotiate a 10-day weather-window extension agreement with Heerema Marine Contractors to prevent standby penalty charges (\$150\$/day\$).

Web Dashboard Integration

The interactive web dashboard at [04_dashboard/index.html](file:///C:/Users/frank/Desktop/EVM/04_dashboard/index.html) and [04_dashboard/drill_tower_web_report.html](file:///C:/Users/frank/Desktop/EVM/04_dashboard/drill_tower_web_report.html) features:

- Red Indicator Audit Table**: Direct table highlighting root causes, actions, and accountable CAMs.
- Interactive Badges & Audit Modal**: Clicking any red status metric badge launches an interactive modal popup with detailed root causes, step-by-step action plans, and accountable owner contact info.
- S-Curve Dynamic Layer Toggles**: Allows switching between Cost Variance (\$CV\$), Schedule Variance (\$SV\$), and Earned Schedule (\$ES\$).

5. Monthly Cash Burn Speed & Budget Depletion (Burn Out)

Project Name: Offshore EPC Platform Drill Tower Project

Status Date: August 31, 2026 (Month 8 / Status Week 36)

Author: Frank Ellingsen, Lead Project Controller

🔗 Executive Burn Rate & Cash Runway Summary

Burn Rate Metric	Historical Value (Months 1-8)	Forecast Value (Months 9-13)	Benchmark Target	Evaluation & Impact
Average Monthly Burn Rate	**\$2,930,000/month**	**\$2,394,721/month**	\$2,208,333/month	**EXPEDITED BURN**. Spending 32.7% faster than baseline monthly plan.
Peak Monthly Spend	**\$4,450,000/month** (Jun)	**\$3,300,000/month** (Oct)	\$4,000,000/month	High yard labor density & vessel standby daily rates.
Cumulative Actual Spend	**\$23,440,000**	**\$35,413,604** (EAC)	\$20,500,000 (PV)	**-\$5.90M Cost Overrun to Date**.
Budget Burn Out Month	**Month 9 (September 2026)**	-	Month 12 (Dec 2026)	**RUNWAY EXHAUSTED**. 100% of baseline budget (\$BAC\$) depleted in Month 9.
Outturn Overrun Cash Gap	**+\$8,913,604**	-	\$0.00	Capital overrun required to reach completion in Month 13.

📊 Monthly Burn Rate & Cumulative Expenditure Table

Month	Date	PV Monthly	EV Monthly	AC Monthly	Cumulative PV	Cumulative EV	Cumulative AC	Budget Status / Burn Out Runway
M1	Jan 31	\$300,000	\$300,000	\$320,000	\$300,000	\$300,000	\$320,000	Baseline Runway (\$26.18M remaining)
M2	Feb 28	\$900,000	\$840,000	\$950,000	\$1,200,000	\$1,140,000	\$1,270,000	Baseline Runway (\$25.23M remaining)
M3	Mar 31	\$2,700,000	\$2,260,000	\$2,900,000	\$3,900,000	\$3,400,000	\$4,170,000	Baseline Runway (\$22.33M remaining)
M4	Apr 30	\$3,800,000	\$3,520,000	\$4,050,000	\$7,700,000	\$6,920,000	\$8,220,000	Baseline Runway (\$18.28M remaining)
M5	May 31	\$3,000,000	\$2,900,000	\$3,220,000	\$10,700,000	\$9,820,000	\$11,440,000	Baseline Runway (\$15.06M remaining)
M6	Jun 30	\$4,000,000	\$2,750,000	\$4,450,000	\$14,700,000	\$12,570,000	\$15,890,000	Baseline Runway (\$10.61M remaining)
M7	Jul 31	\$3,000,000	\$2,620,000	\$3,850,000	\$17,700,000	\$15,190,000	\$19,740,000	Baseline Runway (\$6.76M remaining)
M8	Aug 31	\$2,800,000	\$2,350,000	\$3,700,000	\$20,500,000	\$17,540,000	**\$23,440,000**	**CRITICAL RUNWAY** (\$3.06M remaining)
M9	Sep 30	\$2,500,000	\$2,100,000	**\$3,200,000**	\$23,000,000	\$19,640,000	**\$26,640,000**	✖ **BUDGET BURN OUT** (\$BAC \$26.5M Depleted!)
M10	Oct 31	\$1,500,000	\$2,200,000	**\$3,300,000**	\$24,500,000	\$21,840,000	**\$29,940,000**	Overrun Phase (+\$3.44M Over budget)

Month	Date	PV Monthly	EV Monthly	AC Monthly	Cumulative PV	Cumulative EV	Cumulative AC	Budget Status / Burn Out Runway
M11	Nov 30	\$1,000,000	\$2,000,000	**\$2,500,000**	\$25,500,000	\$23,840,000	**\$32,440,000**	Overrun Phase (+\$5.94M Over budget)
M12	Dec 31	\$1,000,000	\$1,800,000	**\$1,800,000**	\$26,500,000	\$25,640,000	**\$34,240,000**	Overrun Phase (+\$7.74M Over budget)
M13	Jan 31	\$0	\$860,000	**\$1,173,604**	\$26,500,000	\$26,500,000	**\$35,413,604**	**FINAL COD** (\$EAC Outturn Reached)

 Visual Burn Rate Dynamics

CUMULATIVE CASH BURN (\$M)

\$36M +-----* (Final EAC: \$35.41M)

\$32M | *-----' (Overrun Expenditure)

\$28M | *-----'

\$26.5M+-----* BURN OUT----- (BAC Budget Ceiling)

\$24M | *-----'

\$20M | *-----'

\$16M | *-----'

\$12M | *-----'

\$8M | *-----'

\$4M | *-----'

\$0 +--+-----+-----+-----+-----+-----+-----+-----+-----+

M1 M3 M5 M7 M8(Now) M9(Burn) M11 M13(COD)

 Key Project Controller Recommendations

1. ****Secure Overrun Credit Line Before Month 9****:
- The project will exhaust its \$26.50M baseline budget in ****September 2026 (Month 9)****. Emergency liquidity approval of ****\$8.91M**** must be finalized prior to September 15.
2. ****Cap Monthly Burn Speed at Egersund Yard****:
- Monthly actual spend averaged ****\$3.78M/month**** in Months 6-8. Impose a monthly spending limit of ****\$3.00M/month**** for Q4 yard operations.

6. Cost Variance Waterfall Bridge (\$BAC to EAC)

****Project Name**:** Offshore EPC Platform Drill Tower Project

****Status Date**:** August 31, 2026 (Month 8 / Status Week 36)

****Author**:** Frank Ellingsen, Lead Project Controller

Executive Cost Variance Waterfall Bridge Summary

The ****Cost Variance Waterfall Bridge**** reconciles the step-by-step progression from the original ****Baseline Budget (\$BAC = **\$26,500,000**)**** to the final ****Outturn Forecast (\$EAC_1 = **\$35,413,604**)****, isolating the exact financial impact of cost variances (\$VAC\$) across each WBS element and time-based overhead extension.

> **[!IMPORTANT]**

> ****Total Project Variance**:** ****+\$8,913,604**** (+33.6% cost expansion).

> ****Primary Cost Driver**:** Egersund Derrick Mast Assembly Rework (WBS 1.3.2) contributing ****\$2.40M (26.9%)**** of total cost expansion.

> ****Secondary Cost Driver**:** Earned Schedule Time Delay & Extended Overhead (SPI_t = 0.9250) contributing ****\$3.81M (42.8%)**** of outturn escalation.

Step-by-Step Waterfall Cost Bridge Table

Bridge Step	WBS Code	Control Account / Variance Component	Incremental Cost Impact	Cumulative Project Cost	% Contribution to Total Overrun	Primary Root Cause & Description
0	`1.0`	**Original Baseline Budget (\$BAC\$)**	**\$26,500,000**	**\$26,500,000**	-	Original approved EPC contract target budget.
1	`1.1.1`	Detail Structural Engineering	**+\$300,000**	\$26,800,000	3.4%	Redesign of connection plates & third-party DNV class review.
2	`1.2.1`	Procurement & Tubular Steel	**+\$600,000**	\$27,400,000	6.7%	Raw subsea tubular steel price inflation & mud pump shipping surcharge.
3	`1.3.1`	Verdal Sub-Structure Fabrication	**+\$200,000**	\$27,600,000	2.2%	Minor yard overtime for weld inspection.
4	`1.3.2`	**Egersund Derrick Mast Rework**	**+\$2,400,000**	**\$30,000,000**	**26.9%**	**PRIMARY REWORK DRIVER**: Pipe tolerance mismatch requiring 24/7 NDT welding.
5	`1.4.1`	Heavy Lift Vessel Standby	**+\$1,300,000**	\$31,300,000	14.6%	Marine weather standby daily rates (*Heerema Sleipnir*).
6	`1.4.2`	Offshore Topside Lifting & Mating	**+\$100,000**	\$31,400,000	1.1%	Pre-lift trial fitting and alignment fixtures.
7	`1.5.1`	Hook-up & Commissioning	**+\$200,000**	\$31,600,000	2.2%	Extended offshore commissioning specialist crew.
8	`EAC_t`	**Schedule Delay Overhead Extension**	**+\$3,813,604**	**\$35,413,604**	**42.8%**	**TIME DELAY SPREAD**: +31-day schedule delay (\$SPI_t = 0.9250\$) spreading PMO & site overhead.
END	`1.0`	**Final Forecast Outturn (\$EAC\$)**	**+\$8,913,604**	**\$35,413,604**	**100.0%**	Final predicted cost outcome at completion.

Visual Waterfall Cost Progression

...

BASE \$26.50M (BAC)

|

|——[+\$0.30M] Step 1: Detail Engineering Revisions

|

|——[+\$0.60M] Step 2: Tubular Steel Procurement Inflation

|

|——[+\$0.20M] Step 3: Verdal Sub-Structure Fabrication

|

|——[+\$2.40M]  Step 4: Egersund Derrick Mast Yard Rework (MAJOR YARD DRIVER)

|

|——[+\$1.30M]  Step 5: Heavy Lift Vessel Weather Standby

|

|——[+\$0.10M] Step 6: Topside Mating Fit-up

|

|——[+\$0.20M] Step 7: Hook-up Commissioning Crew

|

|——[+\$3.81M]  Step 8: Time Delay Overhead Spread (SCHEDULE IMPACT)

|

OUTTURN \$35.41M (Final EAC)

...

Steering Committee Control Takeaways

1. **Yard Rework Containment**:

- \$WBS\text{ }1.3.2\$ represents \$26.9\%\$ of total overrun. Enforce strict fixed-fee caps on remaining yard welding hours.

2. **Schedule Recovery to Limit Overhead Spread**:

- \$42.8\%\$ of total overrun stems from time-based project management & site overhead extension (\$+\\$3.81\text{M}\$). Recovering 15 calendar days reduces overhead by \$\approx \\$1.80\text{M}\$.

7. EPC Executive Risk Matrix & Mitigation Register

****Project Name**:** Offshore EPC Platform Drill Tower Project

****Status Date**:** August 31, 2026 (Month 8 / Status Week 36)

****Author**:** Frank Ellingsen, Lead Project Controller

 Executive Risk Heatmap Matrix (5x5 Grid)

The 5x5 Risk Heatmap Matrix plots project risks by ****Probability (Likelihood 1-5)**** versus ****Impact (Severity 1-5)****. Total Risk Score = \$Probability \times Impact\$.

...

PROBABILITY

(5) Almost Certain | [5] | [10] | [15] | [20] | [25] |

(4) High | [4] | [8] | [12] |  R02 |  R01 | <-- CRITICAL ZONE

(3) Moderate | [3] | [6] |  R04 |  R03 | [15] |

(2) Low | [2] | [4] |  R05 | [8] | [10] |

(1) Rare | [1] | [2] | [3] | [4] | [5] |

+-----+-----+-----+-----+-----+

(1) (2) (3) (4) (5)

Negligible Minor Moderate Major Critical

IMPACT

...

 Comprehensive Risk Register & Mitigation Action Plan

Risk ID	Risk Category	Risk Event Description	WBS	Risk Owner (CAM)	Prob (1-5)	Imp (1-5)	Risk Score	Risk Exposure (\$)	Actionable Mitigation Strategy	Status
R01	**Fabrication**	**Egersund Mast Assembly Dimensional Out-of-Tolerance & Yard Labor Rework**	1.3.2	O. Eriksen	4	5	20 (Critical)	**\$2,400,000**	Deploy 24/7 dual-shift NDT welding specialists; cap yard labor billing under a fixed-fee agreement.	**Active**
R02	**Offshore Marine**	**Heavy Lift Vessel Mobilization Delay & North Sea Weather Standby**	1.4.1	K. Solberg	4	4	16 (Critical)	**\$1,800,000**	Negotiate flexible weather window with Heerema; monitor 7-day wave height (\$Hs < 2.5m\$).	**Active**
R03	**Procurement**	**High-Grade Subsea Tubular Steel	1.2.1	M. Berg	3	4	12 (High)	**\$1,200,000**	Dual-source tubular steel from	**Monitored**

Risk ID	Risk Category	Risk Event Description	WBS	Risk Owner (CAM)	Prob (1-5)	Imp (1-5)	Risk Score	Risk Exposure (\$)	Actionable Mitigation Strategy	Status
		Mill Delivery Lags & Price Inflation**							European backup mills; arrange expedited hot-shot freight.	
R04	**Offshore Hook-Up**	**Topside Lifting & Mating Mechanical Interface Interference**	1.4.2	T. Nygård	3	3	9 (Medium)	**\$600,000**	Perform 3D laser scan trial fit at Egersund yard prior to offshore mobilization.	**Monitored**
R05	**Engineering**	**Structural Detail Engineering Interface Errors & AFC Revisions**	1.1.1	H. Lindqvist	2	3	6 (Low)	**\$300,000**	Enforce 100% 3D CAD clash detection and third-party DNV class design verification.	**Closed**



Quantitative Financial Risk Summary

- **Total Unmitigated Financial Exposure**: **\$6,300,000**
- **Post-Mitigation Expected Monetary Value (EMV)**: **\$3,410,000**
- **Recommended Management Risk Reserve**: **\$3,500,000** (Aligned with P90 Monte Carlo simulation reserve of $\$401,598$ above $\$35.41\text{M}$ outturn).



Risk Governance & Escalation Triggers

1. **Red Risk Escalation (Score ≥ 15)**: Weekly review with Project Director & CFO. Immediate CAM intervention required.
2. **Amber Risk Monitoring (Score $8 - 14$)**: Bi-weekly review at Project Control Meetings.
3. **Green Risk Tracking (Score ≤ 7)**: Monthly review by CAMs.

8. Vertical Swimlane WBS Control Account Breakdown

Project Name: Offshore EPC Platform Drill Tower Project

Status Date: August 31, 2026 (Month 8 / Status Week 36)

Author: Frank Ellingsen, Lead Project Controller

Executive Category Breakdown Summary

Category / Discipline	WBS Code	Control Account Manager (CAM)	Baseline Budget (\$BAC\$)	Outturn Forecast (\$EAC\$)	Cost Variance (\$VAC\$)	Physical Progress (%)	Current Status
1. Engineering & Design	`1.1`	H. Lindqvist	**\$3,200,000**	**\$3,500,000**	-\$300,000	**100%**	Achieved
2. Procurement & Supply Chain	`1.2`	M. Berg	**\$8,500,000**	**\$9,100,000**	-\$600,000	**94.1%**	Advanced
3. Yard Fabrication & Assembly	`1.3`	O. Eriksen	**\$9,800,000**	**\$12,400,000**	-\$2,600,000	**77.6%**	Critical Yard Rework
4. Offshore Installation & Marine	`1.4`	K. Solberg	**\$5,000,000**	**\$6,400,000**	-\$1,400,000	**19.2%**	Standby Risk
5. Hook-up & Commissioning	`1.5`	T. Nygård	**\$1,800,000**	**\$2,000,000**	-\$200,000	**0.0%**	Scheduled Q4
TOTAL PROJECT	`1.0`	**Frank Ellingsen**	**\$26,500,000**	**\$35,413,604**	**-\$8,913,604**	**66.2%**	**Overrun & Delayed**

Vertical Swimlane Work Breakdown Structure (WBS)

+-----+ +-----+ +-----+ +-----+ +-----+									
1.1 ENGINEERING		1.2 PROCUREMENT		1.3 FABRICATION		1.4 INSTALLATION		1.5 COMMISSIONING	
+-----+ +-----+ +-----+ +-----+ +-----+									
1.1.1 Detail Eng		1.2.1 Steel Pipes		1.3.1 Verdal Sub	1.4.1 Heavy Lift		1.5.1 Hook-up & NDT		
BAC: \$1.80M	100%	BAC: \$5.00M	100%	BAC: \$5.00M	90%	BAC: \$3.20M	30%	BAC: \$1.00M	0%
EAC: \$2.10M (-\$300K)		EAC: \$5.50M (-\$500K)		EAC: \$5.20M (-\$200K)		EAC: \$4.50M (-\$1.3M)		EAC: \$1.10M (-\$100K)	
+-----+ +-----+ +-----+ +-----+ +-----+									
1.1.2 Piping Design		1.2.2 Mud Pumps		1.3.2 Egersund Mast	1.4.2 Topside Mating		1.5.2 Handover COD		
BAC: \$1.40M	100%	BAC: \$3.50M	85%	BAC: \$4.80M	65%	BAC: \$1.80M	0%	BAC: \$0.80M	0%

1.1.2 Piping Design	1.2.2 Mud Pumps	1.3.2 Egersund Mast	1.4.2 Topside Mating	1.5.2 Handover COD
EAC: \$1.40M (\$0K)	EAC: \$3.60M (-\$100K)	EAC: \$7.20M (-\$2.4M)	EAC: \$1.90M (-\$100K)	EAC: \$0.90M (-\$100K)

+-----+ +-----+ +-----+ +-----+ +-----+

...

 Discipline Performance Insights

1. **Engineering Swimlane (\$1.1\$)**: Completed 100% physical progress with a minor $-\$300\text{K}$ cost variance due to design revisions on structural steel connection plates.
2. **Procurement Swimlane (\$1.2\$)**: 94.1% complete. Steel tubular procurement encountered $-\$500\text{K}$ raw material price inflation.
3. **Fabrication Swimlane (\$1.3\$)**: Primary project cost overrun center ($-\$2.60\text{M}$ variance). Verdal yard sub-structure is 90% complete, while Egersund mast assembly requires extended NDT rework ($-\$2.40\text{M}$ overrun).
4. **Installation Swimlane (\$1.4\$)**: Heavy lift vessel mobilization has incurred $-\$1.30\text{M}$ standby cost escalation due to North Sea autumn weather windows.
5. **Commissioning Swimlane (\$1.5\$)**: Scheduled for Q4 2026 / Q1 2027 following completion of topside mating.

9. Project Financial Metrics & Appraisal

Project Name: Offshore EPC Platform Drill Tower Project

CAPEX Baseline (\$BAC): \$26,500,000

CAPEX Outturn Forecast (\$EAC): \$35,413,604

Discount Rate / Hurdle Rate (\$WACC): 10.0%

Author: Frank Ellingsen, Financial Controller & BI Specialist

Executive Summary & Financial Ratio Card

Capital Appraisal Metric	Standard Formula	Outturn Value	Threshold Target	Commercial Decision & Evaluation
Net Present Value (\$NPV)	$\sum \frac{CF_t}{(1+r)^t} - CAPEX$	+\$14,899,563	> \$0.00	ACCEPT . Project creates \$14.90M net shareholder value above 10% WACC.
Internal Rate of Return (\$IRR)	$NPV(IRR) = 0$	18.86%	> 10.0% WACC	EXCEEDS HURDLE . Premium spread of +886 bps over corporate cost of capital.
Profitability Index (\$PI)	$\frac{PV(\text{Inflows})}{CAPEX}$	1.42	> 1.00	STRONG BENCHMARK . Generates \$1.42 in present value per \$1.00 spent.
Simple Payback Period	$\frac{CAPEX}{\text{Annual CF}}$	4.43 Years	< 5.00 Years	VIABLE . Capital recovered in 53.1 months (May 2031).
Total Cumulative ROI	$\frac{\text{Net Cash Flow}}{CAPEX}$	134.37%	> 50.0%	EXCELLENT . Total net cash inflows of \$75.00M on \$35.41M investment.
Annualized ROI (CAGR)	$(1 + ROI)^{1/N} - 1$	8.89%/year	> 7.0%/year	STABLE . Sustained annual compounded return across 10-year asset life.
Future Value (FV at Yr 10)	$\sum CF_t \cdot (1+r)^{(10-t)}$	\$130,499,397	-	Gross compounded terminal cash value generated by Year 10 (2036).
Net Future Value (\$NFV)	$FV(\text{Inflows}) - FV(CAPEX)$	+\$38,645,628	> \$0.00	Net value created in Year 10 dollar terms.

1. Cash Flow Schedule & Capital Outlay

Capital Expenditure (\$CAPEX)

- Original Baseline (\$BAC):** \$26,500,000
- Outturn Forecast (\$EAC₁):** \$35,413,604 (Includes \$8.91M cost overrun from Egersund yard rework and Heerema vessel standby).

Operational Inflows (\$Years_t - 10)

- Daily Rig Revenue Addition:** \$110,000/day × 365 × 85% utilization = \$34,127,500/year
- Annual Operational Expenditure (\$OPEX):** \$26,127,500/year
- Net Annual Cash Flow (\$CF_t):** \$8,000,000/year
- Terminal Salvage Value (Year 10):** \$3,000,000

2. Sensitivity Analysis (Cost Overrun Impact on \$NPV & \$IRR)

CAPEX Scenario	Total Outturn (\$EAC\$)	Net Present Value (\$NPV\$)	Internal Rate of Return (\$IRR\$)	Payback Period	Profitability Index (\$PI\$)
Baseline Plan (\$BAC\$)	\$\text{\\$26.50}\text{(M)}\$	**+\$23,813,167**	**27.42%**	**3.31 Years**	**1.90**
Best Case (\$EAC_2\$)	\$\text{\\$32.40}\text{(M)}\$	**+\$17,913,167**	**21.28%**	**4.05 Years**	**1.55**
Most Likely (\$EAC_1\$)	**\$\text{\\$35.41}\text{(M)}\$**	**+\$14,899,563**	**18.86%**	**4.43 Years**	**1.42**
Worst Case (\$EAC_3\$)	\$\text{\\$37.43}\text{(M)}\$	**+\$12,879,645**	**17.38%**	**4.68 Years**	**1.34**

> [!NOTE]

> Even under the Worst Case Cost Overrun scenario (\$EAC_3 = \$\text{\\$37.43}\text{(M)}\$), the project remains robustly profitable with an **\$NPV\$ of +\$12.88M** and an **\$IRR\$ of 17.38%**, well clear of the **10.0% corporate hurdle rate**.